Exercises: Gradient Descent Convergence

Mathematical Foundations for Machine Learning

Descent Lemma and Basic Convergence

Exercise 1.1 [Easy]
Let $f : \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable with L-Lipschitz continuous gradient, i.e., $\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2$ for all $x, y \in \mathbb{R}^d$.

(a) State the descent lemma: give an upper bound on $f(y)$ in terms of $f(x)$, $\nabla f(x)$, and $\|y - x\|_2^2$.

(b) Using part (a) with $y = x - t\nabla f(x)$, show that for $0 < t \leq \frac{1}{L}$:

$$f(x - t\nabla f(x)) \leq f(x) - \frac{t}{2}\|\nabla f(x)\|_2^2.$$

(c) Explain why this implies that gradient descent with stepsize $t \leq \frac{1}{L}$ produces a monotonically decreasing sequence of objective values.

Exercise 1.2 [Medium] curve main theorem -> abs. value -> + taylor -> yx=f(x) -> n < 2/L

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex with L-Lipschitz continuous gradient, and let x^* be a minimizer with $p^* = f(x^*)$. Consider gradient descent with constant stepsize $t = \frac{1}{L}$:

$$x^{(k+1)} = x^{(k)} - t\nabla f(x^{(k)}).$$ Important use of firsrt-order convexity

(a) Using the first-order convexity condition and the descent lemma, show that:

$$f(x^{(k)} - t\nabla f(x^{(k)})) \leq f(x^*) + \frac{1}{2t}\left(\|x^{(k)} - x^*\|_2^2 - \|x^{(k+1)} - x^*\|_2^2\right).$$

Hint: Write the right-hand side as $f(x^*) + \langle \nabla f(x^{(k)}), x^{(k)} - x^* \rangle - \frac{t}{2}\|\nabla f(x^{(k)})\|_2^2$ *and use* $2\langle a, b\rangle = \|a\|^2 + \|b\|^2 - \|a - b\|^2$.

(b) By averaging over $k = 1, \ldots, K$ and using the telescoping property, derive the convergence rate:

$$f(x^{(K)}) - p^* \leq \frac{\|x^{(0)} - x^*\|_2^2}{2Kt}.$$

sum over 0, k-1, apply telescoping
for upper bound and take biggest o

(c) How many iterations are needed to achieve $f(x^{(K)}) - p^* \leq \varepsilon$? Express your answer in terms of L, $\|x^{(0)} - x^*\|_2$, and ε.

Exercise 1.3 [Hard] Descent lemma continuation -> n = 1/L, take sum, divide by k and take avg -> min

Let $f : \mathbb{R}^d \to \mathbb{R}$ be continuously differentiable (**not necessarily convex**) with L-Lipschitz continuous gradient and bounded below by f^*.

(a) Using the descent lemma, show that for gradient descent with stepsize $t = \frac{1}{L}$:

$$\|\nabla f(x^{(k)})\|_2^2 \leq 2L\left(f(x^{(k)}) - f(x^{(k+1)})\right).$$

(b) By summing from $k = 0$ to K and using the telescoping property, prove:

$$\min_{k=0,\ldots,K} \|\nabla f(x^{(k)})\|_2^2 \leq \frac{2L(f(x^{(0)}) - f^*)}{K + 1}.$$

(c) What convergence guarantee does part (b) give? Why does this result use $\min_k$ instead of $\|\nabla f(x^{(K)})\|^2$? Explain the difference to the convex case.

(d) In the context of training a neural network with a non-convex loss, what does this result tell us about the practical behavior of gradient descent?

Strong Convexity and Linear Convergence

Exercise 2.1 [Easy]
A function $f : \mathbb{R}^d \to \mathbb{R}$ is m-strongly convex if $g(x) = f(x) - \frac{m}{2}\|x\|^2$ is convex.

(a) Show that if f is twice continuously differentiable, then f is m-strongly convex if and only if $\nabla^2 f(x) \succeq mI$ for all x.

(b) Determine whether $f(x) = \frac{1}{2}x^T Ax - b^T x$ is strongly convex, where $A \in \mathbb{R}^{d\times d}$ is symmetric positive definite. If so, find the strong convexity parameter.

(c) Is $f(x) = \|x\|_1$ strongly convex? Justify your answer.

Exercise 2.2 [Medium]
Let $f : \mathbb{R}^d \to \mathbb{R}$ satisfy the Polyak-Łojasiewicz (PL) condition with parameter $\mu > 0$:

$$\frac{1}{2}\|\nabla f(x)\|_2^2 \geq \mu(f(x) - p^*) \quad \forall x \in \mathbb{R}^d,$$

where $p^* = \min_x f(x)$. Assume f has L-Lipschitz continuous gradient.

(a) Show that any m-strongly convex function satisfies the PL condition with $\mu = m$.
Hint: For strongly convex f, minimize $f(y) \geq f(x) + \langle \nabla f(x), y - x\rangle + \frac{m}{2}\|y - x\|^2$ over y.

(b) Using the descent lemma with stepsize $t = \frac{1}{L}$ and the PL condition, show:

$$f(x^{(k+1)}) - p^* \leq \left(1 - \frac{\mu}{L}\right)(f(x^{(k)}) - p^*).$$

(c) Why is the PL condition strictly weaker than strong convexity? Give an example of a function that satisfies PL but is not strongly convex.

Exercise 2.3 [Hard]
Consider $f : \mathbb{R}^d \to \mathbb{R}$ that is m-strongly convex with L-Lipschitz continuous gradient. Define the condition number $\kappa = \frac{L}{m}$.

(a) Using the linear convergence result $f(x^{(k)}) - p^* \leq \left(1 - \frac{1}{\kappa}\right)^k (f(x^{(0)}) - p^*)$, show that gradient descent requires $K = O\left(\kappa \log \frac{1}{\varepsilon}\right)$ iterations to achieve $f(x^{(K)}) - p^* \leq \varepsilon$.
Hint: Use $\log(1 - \frac{1}{\kappa}) \leq -\frac{1}{\kappa}$.

(b) Accelerated gradient descent achieves the rate $\left(1 - \frac{1}{\sqrt{\kappa}}\right)$-linear convergence. How many iterations does it need? Compare to part (a) when $\kappa = 1000$.

(c) Consider Ridge regression: $f(\theta) = \frac{1}{2n}\|X\theta - y\|^2 + \frac{\lambda}{2}\|\theta\|^2$ where $X \in \mathbb{R}^{n\times d}$. Compute the strong convexity parameter m and the Lipschitz constant L of the gradient. Express the condition number in terms of the singular values of X and λ.

(d) For the Ridge regression in (c), explain how increasing λ affects the condition number and hence the convergence speed. Discuss the trade-off with model accuracy.

Stochastic Gradient Descent

Exercise 3.1 [Easy]
Consider the finite-sum objective $F(x) = \frac{1}{n}\sum_{i=1}^n f_i(x)$ and the SGD update $x^{(k+1)} = x^{(k)} - \alpha_k \nabla f_{i_k}(x^{(k)})$, where i_k is drawn uniformly at random from $\{1, \ldots, n\}$.

(a) Show that $\nabla f_{i_k}(x^{(k)})$ is an unbiased estimator of $\nabla F(x^{(k)})$, i.e., $\mathbb{E}[\nabla f_{i_k}(x^{(k)}) \mid x^{(k)}] = \nabla F(x^{(k)})$.

(b) If we use a mini-batch $B \subset \{1, \ldots, n\}$ of size $|B|$ drawn uniformly at random, and define $g_B(x) = \frac{1}{|B|}\sum_{j\in B} \nabla f_j(x)$, show that g_B is also an unbiased estimator of $\nabla F(x)$.

(c) Does increasing the mini-batch size $|B|$ change the bias? What does it change?

Exercise 3.2 [Medium]
Let $F(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$ where ∇F is L-Lipschitz continuous, F is bounded below by F^*, and $\mathbb{E}[\|\nabla f_i(x)\|_2^2] \leq \sigma^2$ for all x. Consider SGD with constant stepsize α.

(a) Show the key SGD descent inequality: for $x^{(k+1)} = x^{(k)} - \alpha g(x^{(k)}, \xi_k)$,

$$\mathbb{E}[F(x^{(k+1)}) \mid x^{(k)}] \leq F(x^{(k)}) - \alpha\|\nabla F(x^{(k)})\|_2^2 + \frac{\alpha^2 L}{2}\sigma^2.$$

(b) Compare this to the deterministic gradient descent bound. What is the additional error term and what causes it?

(c) If F is additionally μ-strongly convex, show that with constant stepsize $\alpha \leq \frac{1}{2L}$:

$$\mathbb{E}[F(x^{(k+1)})] - p^* \leq (1 - 2\alpha\mu)^k (F(x^{(0)}) - p^*) + \frac{\alpha L \sigma^2}{4\mu}.$$

Hint: First show that $(1 - 2\alpha\mu) \in (0, 1)$*, then iterate the one-step bound and use* $\sum_{s=0}^{\infty}(1 - 2\alpha\mu)^s = \frac{1}{2\alpha\mu}$.

Exercise 3.3 [Hard]
Consider SGD for the strongly convex case from the previous exercise, with the convergence bound:

$$\mathbb{E}[F(x^{(k)})] - p^* \leq (1 - 2\alpha\mu)^k (F(x^{(0)}) - p^*) + \frac{\alpha L \sigma^2}{4\mu}.$$

(a) The bound has two terms. Interpret each term: what does it depend on, and what happens to it as $k \to \infty$?

(b) Consider the stepsize halving strategy: run SGD until $\mathbb{E}[F(x^{(k)})] - p^* \leq \frac{\alpha L \sigma^2}{2\mu}$, then halve α and repeat. Explain why this is justified by the convergence bound.

(c) Compare the computational cost per iteration of SGD vs. full gradient descent on the finite-sum objective $F(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$. In a regime where n is very large and moderate accuracy suffices, argue why SGD is preferred despite its inferior convergence rate.

(d) In the context of training a deep neural network with cross-entropy loss, the function F is non-convex. Discuss whether the convergence guarantees above still apply. What weaker guarantees can we state?

Convergence Rate Comparisons and Stepsize Schedules

Exercise 4.1 [Easy]
The convergence rate of SGD in the non-convex case depends on the stepsize schedule through the ratio $\frac{\sum_{k=1}^{t} \alpha_k^2}{\sum_{k=1}^{t} \alpha_k}$.

(a) Compute this ratio for constant stepsize $\alpha_k = \alpha$.

(b) Compute (or bound) this ratio for $\alpha_k = \frac{\alpha}{k}$. Use $\sum_{k=1}^{t} \frac{1}{k} \approx \log t$ and $\sum_{k=1}^{t} \frac{1}{k^2} \leq \frac{\pi^2}{6}$.

(c) Compute (or bound) this ratio for $\alpha_k = \frac{\alpha}{\sqrt{k}}$. Use $\sum_{k=1}^{t} \frac{1}{\sqrt{k}} \approx 2\sqrt{t}$ and $\sum_{k=1}^{t} \frac{1}{k} \approx \log t$.

(d) Which schedule gives the best convergence rate for large t? Rank them.

Exercise 4.2 [Medium]
Consider the following table of convergence rates for unconstrained minimization of a function with L-Lipschitz continuous gradient:

Setting	**GD rate**	**Accelerated GD rate**
Convex	$O(L/k)$	$O(L/k^2)$
Strongly convex ($m > 0$)	$(1 - m/L)$-linear	$(1 - \sqrt{m/L})$-linear

(a) For the convex case, how many iterations does GD need to achieve ε-accuracy? How many does accelerated GD need? Express in terms of L and ε.

(b) For the strongly convex case, how many iterations does GD need to achieve ε-accuracy? How many does accelerated GD need? Express in terms of $\kappa = L/m$ and ε.

(c) Are the accelerated rates optimal? State what is known about lower bounds.

(d) A convex function f has $L = 100$ and $m = 0$ (convex but not strongly convex). Another function g has $L = 100$ and $m = 1$. For target accuracy $\varepsilon = 10^{-6}$, compare the number of GD iterations needed for each. What does this tell us about the importance of strong convexity?

Exercise 4.3 [Hard]
(Backtracking line search and convergence.) Consider gradient descent with backtracking line search: starting from $t = t_{\max}$, we repeatedly set $t \leftarrow \beta t$ (with $0 < \beta < 1$) until the Armijo condition is satisfied:

$$f(x^{(k)} - t\nabla f(x^{(k)})) \leq f(x^{(k)}) + \langle \nabla f(x^{(k)}), -t\nabla f(x^{(k)})\rangle + \frac{1}{2t}\|t\nabla f(x^{(k)})\|^2.$$

(a) Simplify the Armijo condition above. Show it is equivalent to:

$$f(x^+) \leq f(x^{(k)}) - \frac{t}{2}\|\nabla f(x^{(k)})\|^2.$$

(b) Explain why the backtracking line search always terminates if ∇f is L-Lipschitz continuous. What is the smallest stepsize that is guaranteed to be accepted?

(c) Let $t_{\min} = \min_{k=0,\ldots,K} t_k$ be the smallest stepsize used. Show that the convergence bound for gradient descent with constant stepsize t still holds when replacing t with $t_{\min}$.

(d) In practice, backtracking line search is often more efficient than using the fixed stepsize $t = 1/L$. Explain why, considering that L might be a loose global bound.

Mixed Challenging Exercises

Exercise 5.1 [Hard]
(From theory to practice: learning rate warmup.) Consider SGD on a μ-strongly convex function with L-Lipschitz gradient. The convergence bound with constant stepsize α is:

$$\mathbb{E}[F(x^{(k)})] - p^* \leq (1 - 2\alpha\mu)^k (F(x^{(0)}) - p^*) + \frac{\alpha L \sigma^2}{4\mu}.$$

(a) What is the optimal constant stepsize α^* that minimizes the asymptotic error floor $\frac{\alpha L\sigma^2}{4\mu}$ subject to maintaining stability (i.e., $(1 - 2\alpha\mu) \in (0, 1)$, which requires $\alpha < \frac{1}{2\mu}$)? Explain the trade-off.

(b) In practice, the strategy is to start with a small α, increase it (warmup), then decrease it. Using the bound, explain why starting with a very large α can be problematic. What does the contraction factor $(1 - 2\alpha\mu)$ become if $\alpha > \frac{1}{2\mu}$?

(c) Consider a two-phase strategy: Phase 1 uses $\alpha_1 = \frac{1}{2L}$ for K_1 iterations, then Phase 2 uses $\alpha_2 = \frac{1}{2L} \cdot \frac{1}{10}$ for K_2 iterations. What is the noise floor in each phase? If $\kappa = 100$ and $\sigma^2 = 1$, compute the noise floors numerically.

(d) Nesterov's accelerated gradient descent achieves $(1 - \sqrt{m/L})$-linear convergence but applies only to deterministic gradients. Explain why the momentum term $\frac{k}{k+3}(x^{(k+1)} - x^{(k)})$

would amplify noise in the stochastic setting. Why is the sequence $\{f(x^{(k)})\}$ not monotonically decreasing for accelerated GD?

Exercise 5.2 [Hard]
(Convergence landscape analysis.) Let $F(x) = \frac{1}{n}\sum_{i=1}^{n} f_i(x)$ where each $f_i : \mathbb{R}^d \to \mathbb{R}$ is differentiable.

(a) The gradient variance at a point x is defined as $V(x) = \frac{1}{n}\sum_{i=1}^{n} \|\nabla f_i(x) - \nabla F(x)\|^2$. Show that $V(x^*) = 0$ if and only if $\nabla f_i(x^*) = \nabla F(x^*) = 0$ for all i (i.e., all component functions share the same minimizer).

(b) In practice, for overparameterized neural networks on a training set, $V(x^*) \approx 0$ near a good solution. Explain how this connects to the phenomenon that SGD can converge faster than the general theory predicts.

(c) Consider the special case $f_i(x) = \frac{1}{2}(a_i^T x - b_i)^2$ (least squares). Compute $\nabla f_i(x)$, $\nabla F(x)$, and the gradient variance $V(x)$ at the minimizer $x^* = (A^T A)^{-1} A^T b$ where A has rows a_i^T. Under what condition is $V(x^*) = 0$?

(d) Suppose F satisfies the PL condition with parameter μ and the gradient variance is bounded by $V(x) \leq \rho(F(x) - p^*)$ for some $\rho > 0$ (variance vanishes near the optimum). Using the SGD descent inequality, argue informally that SGD with appropriate stepsize can achieve linear convergence *without* a noise floor.